MIL-C-48411/7A(MU)
1 December 1975
SUPERSEDING
MIL-C-48411/7(MU)
25 April 1974

MILITARY SPECIFICATION
CIRCUIT CARD ASSEMBLY: 11746264
TIMING-3

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-48411 comprise the complete requirements for Timing-3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

11746264	Circuit Card Assembly, Timing-3

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item identification.- This assembly is the third of three (3) single ended printed circuit board consisting of counters and digital-to-analog converters necessary to provide the analog voltages necessary to be used in the analog computer section of the STG.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11746264

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

3.2 Characteristics.

3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power Supply.- The requirements of 3.2 shall be met with the following supply voltages.

a. +5 VDC
b. +15 VDC
c. -15 VDC

3.2.1.2 Timing Networks.- The outputs on pins 32 and 14 shall be capable of exhibiting a voltage of from 0 to ± 10 VDC with a 5 mV resolution per input pulse.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on drawing 11746264 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-48411, Drawing 11746264 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-C-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-C-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-48411.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-48411.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

4.2.1 Test equipment. (see Figure 1)

4.2.1.1 Power supply (PS1).- The power supply shall be 10 ± 0.5 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.2 Power supply (PS2 & PS3).- The power supply shall be 20 ± 1.0 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.3 Dummy Loads.- The dummy loads shall be two (2) 10K Ohm ± 5%, 1/4 watt resistors.

4.2.1.4 Oscilloscope.- Rise time of scope plus associated measuring probe of 50 nanoseconds or less, sweep period of 50 nanoseconds per centimeter of greater, input resistance greater than 8 megohms, input capacitance less than 25 picafarads.

4.2.1.5 Pulse Generator.- The generator shall be capable of producing 60 ± 1 Hz clock pulse and a 10 ± 1 mS, 5 VDC pulse.

4.2.1.6 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these tests shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power supply.- All the requirements shall be met with the following power supply voltages.

a. +5 ± 0.05 VDC
b. +15 ± 0.05 VDC
c. -15 ± 0.05 VDC

4.2.2.2 Digital Output.- The "0" or false output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between +2.40 and +5.00 VDC.

4.2.2.3 Test Set-up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.4 Timing network.

4.2.2.4.1 Timing network-1.- With Logic "1" applied to pin 30 and Logic "0" applied to pin 23, the output at pin 32 shall be 0 ± 50 mVDC.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

By applying a 10ns, 5V pulse to pin 28, the output on pin 32 shall be decreased by 5 ± 1 mV. For each additional pulse applied, the output shall be correspondingly reduced. Remove the 10 mS pulse source from pin 28 and apply a 60 Hz clock pulse which shall cause the output at pin 32 to incrementally negatively increased to -10.00 ± 0.25 VDC then start at +10.00 ± 0.25 VDC and decrease to 0.00. Repeat the above procedure with Logic "1" applied to pin 28 and the pulses to pin 30 which will cause the outputs at pin 32 to be positive going.

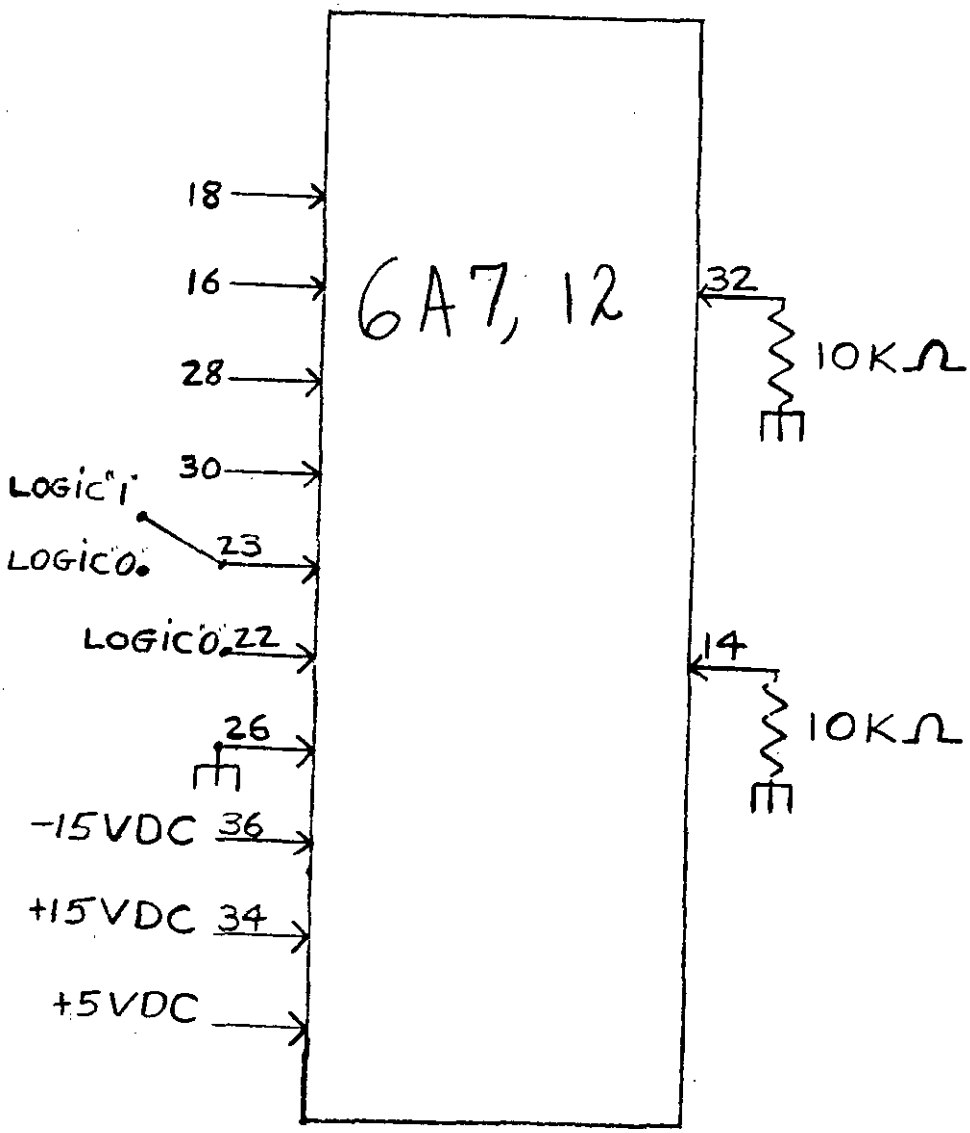


FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

4.2.2.4.2 Timing network-2.- With Logic "1" applied to pin 16 and Logic "0" applied to pin 23, the output at pin 14 shall be 0 ± 50 mVDC. By applying a 10mS , 5V pulse to pin 18, the output on pin 14 shall be decreased by 5 ± 1 MV. For each additional pulse applied, the output shall be correspondingly reduced. Remove the 10 mS pulse source from pin 18 and apply a 60 Hz clock pulse which shall cause the output at pin 14 to incrementally negatively increased to -10.00 ± 0.25 VDC then start at +10.00 ± 0.25 VDC and decrease to 0.00. Repeat the above procedure with Logic "1" applied to pin 18 and the pulses to pin 16 which will cause the outputs to be positive going.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11746264.

6. NOTES

6.1 The notes specified in the general specification MIL-C-48411 are applied to this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1240-A602

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-C-48411/7A(MU) Circuit Card Assembly: 11746264 Timing-3

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Frankford Arsenal
ATTN: SARFA-MDM
Phila., PA 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.